

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>
Distance covered while driving to the turnaround point = speed * time = 60 * 3 = 180 miles
Remaining distance to home = 180 miles
Time spent in stand still traffic = 2 hours
Time spent driving at 30 mph = 0.5 hours
Remaining time to reach home = 4 - 2 - 0.5 = 1.5 hours
Distance covered at 30 mph = speed * time = 30 * 0.5 = 15 miles
Remaining distance to home after driving at 30 mph = 180 - 15 = 165 miles
Distance covered at 80 mph = speed * time = 80 * 1.5 = 120 miles
Remaining distance to home after driving at 80 mph = 165 - 120 = 45 miles
So , John is 45 miles from home at the end of the 4 hours .
</think><answer>
45 </answer>

